

Supplementary Information for
MSGNet: sequence learning enhanced with multi-scale De Bruijn graphs
for transcription factor binding site prediction

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Supplementary sections

Section S1. Evaluation metrics

In this work, the prediction of transcription factor binding sites (TFBSs) is formulated as a binary classification task. To comprehensively evaluate model performance, we employ accuracy (Accuracy), the area under the receiver operating characteristic curve (ROC-AUC), and the area under the precision-recall curve (PR-AUC) as primary evaluation metrics.

First, Accuracy is used to measure the overall correctness of the model's predictions, computed as follows:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where TP denotes the number of true positives, TN the number of true negatives, FP the number of false positives, and FN the number of false negatives. However, in highly imbalanced settings (e.g., when negative samples far outnumber positive ones), accuracy can be misleading and fail to reflect the model's ability to detect critical positive cases.

To improve evaluation reliability, we incorporate the receiver operating characteristic (ROC) curve and its area under the curve (ROC-AUC). The ROC curve consists of the false positive rate (FPR) and true positive rate (TPR) evaluated across varying decision thresholds, defined as follows:

$$(FPR, TPR) = \left(\frac{FP}{FP + TN}, \frac{TP}{TP + FN} \right) \quad (2)$$

where FPR denotes the false positive rate, and TPR denotes the true positive rate. The ROC-AUC ranges from 0 to 1, with values closer to 1 indicating stronger discriminative ability of the model.

However, ROC-AUC may still be influenced by the majority class in highly imbalanced scenarios, so we introduce the precision-recall (PR) curve and its area (PR-AUC) as complementary metrics. The PR curve plots recall against precision, defined as follows:

$$(Recall, Precision) = \left(\frac{TP}{TP + FN}, \frac{TP}{TP + FP} \right) \quad (3)$$

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

PR-AUC provides a more sensitive measure of performance on the minority class (i.e., positive TFBS samples) and is more appropriate for assessing models on imbalanced datasets.

Supplementary Tables

Table S1: The statistics of the 165 ChIP-seq datasets

TFs	Motif_set	Train_set	Test_set
CEBPB	wgEncodeAwgTfbsHaibHepg2Cebpbsc150V0416101UniPk	28491	7123
	wgEncodeAwgTfbsHaibK562Cebpbsc150V0422111UniPk	34747	8687
	wgEncodeAwgTfbsSydhA549CebpIggrabUniPk	60745	15187
	wgEncodeAwgTfbsSydhH1hesCebpIggrabUniPk	24472	6118
	wgEncodeAwgTfbsSydhHelas3CebpIggrabUniPk	94892	23724
	wgEncodeAwgTfbsSydhHepg2CebpForsklnUniPk	29012	7254
	wgEncodeAwgTfbsSydhHepg2CebpIggrabUniPk	89115	22279
	wgEncodeAwgTfbsSydhImr90CebpIggrabUniPk	109712	27428
COREST	wgEncodeAwgTfbsSydhK562CebpIggrabUniPk	60416	15104
	wgEncodeAwgTfbsSydhHepg2Corestsc30189IggrabUniPk	8515	2129
CTCF	wgEncodeAwgTfbsSydhK562Corestsc30189IggrabUniPk	56211	14053
	wgEncodeAwgTfbsBroadDnd41CtcfUniPk	77531	19383
	wgEncodeAwgTfbsBroadGm12878CtcfUniPk	69435	17359
	wgEncodeAwgTfbsBroadH1hesCtcfUniPk	103174	25794
	wgEncodeAwgTfbsBroadHelas3CtcfUniPk	81049	20263
	wgEncodeAwgTfbsBroadHepg2CtcfUniPk	72324	18082
	wgEncodeAwgTfbsBroadHmecCtcfUniPk	61801	15451
	wgEncodeAwgTfbsBroadHsmmCtcfUniPk	79036	19760
	wgEncodeAwgTfbsBroadHsmtCtcfUniPk	74404	18602
	wgEncodeAwgTfbsBroadHuvecCtcfUniPk	58102	14526
	wgEncodeAwgTfbsBroadK562CtcfUniPk	80417	20105
	wgEncodeAwgTfbsBroadNhaCtcfUniPk	59652	14914
	wgEncodeAwgTfbsBroadNhdfadCtcfUniPk	75851	18963
	wgEncodeAwgTfbsBroadNhekCtcfUniPk	72955	18239
	wgEncodeAwgTfbsBroadNhlfCtcfUniPk	61337	15335
	wgEncodeAwgTfbsBroadOsteoblCtcfUniPk	86148	21538
	wgEncodeAwgTfbsUtaGm12878CtcfUniPk	75536	18884
	wgEncodeAwgTfbsUtaH1hesCtcfUniPk	67601	16901
	wgEncodeAwgTfbsUtaHelas3CtcfUniPk	90883	22721
	wgEncodeAwgTfbsUtaK562CtcfUniPk	86808	21702
	wgEncodeAwgTfbsUtaMcf7CtcfUniPk	90049	22513
ELF1	wgEncodeAwgTfbsHaibA549Elf1V0422111Etoh02UniPk	13468	3368
	wgEncodeAwgTfbsHaibGm12878Elf1sc631V0416101UniPk	35684	8922
	wgEncodeAwgTfbsHaibHepg2Elf1sc631V0416101UniPk	28315	7079
	wgEncodeAwgTfbsHaibK562Elf1sc631V0416102UniPk	43403	10851
EZH2	wgEncodeAwgTfbsBroadDnd41Ezh239875UniPk	2681	671
	wgEncodeAwgTfbsBroadHelas3Ezh239875UniPk	2840	710
	wgEncodeAwgTfbsBroadHepg2Ezh239875UniPk	5235	1309
	wgEncodeAwgTfbsBroadHsmmEzh239875UniPk	2464	616
	wgEncodeAwgTfbsBroadHuvecEzh239875UniPk	6788	1698
	wgEncodeAwgTfbsBroadK562Ezh239875UniPk	2684	672
FOS	wgEncodeAwgTfbsHaibA549Fosl2V0422111Etoh02UniPk	44750	11188
	wgEncodeAwgTfbsHaibH1hesFosl1sc183V0416102UniPk	1758	440
	wgEncodeAwgTfbsHaibHepg2Fosl2V0416101UniPk	39678	9920

	wgEncodeAwgTfbsHaibK562Fosl1sc183V0416101UniPk	17569	4393
	wgEncodeAwgTfbsSydhGm12878CfosUniPk	3488	872
	wgEncodeAwgTfbsSydhHelas3CfosUniPk	14494	3624
	wgEncodeAwgTfbsSydhHuvecCfosUcdUniPk	72961	18241
	wgEncodeAwgTfbsSydhK562CfosUniPk	11968	2992
	wgEncodeAwgTfbsSydhMcf10aesCfosTam112hHvdUniPk	143425	35857
	wgEncodeAwgTfbsSydhMcf10aesCfosTam14hHvdUniPk	134080	33520
	wgEncodeAwgTfbsSydhMcf10aesCfosTamHvdUniPk	109454	27364
	wgEncodeAwgTfbsUchicagoK562EfesUniPk	16120	4030
FOXA1	wgEncodeAwgTfbsHaibA549Foxa1V0416102Dex100nmUniPk	12054	3014
	wgEncodeAwgTfbsHaibEcc1Foxa1sc6553V0416102Dm002p1hUniPk	9283	2321
	wgEncodeAwgTfbsHaibHepg2Foxa1sc101058V0416101UniPk	67550	16888
	wgEncodeAwgTfbsHaibHepg2Foxa1sc6553V0416101UniPk	80224	20056
	wgEncodeAwgTfbsHaibT47dFoxa1sc6553V0416102Dm002p1hUniPk	65257	16315
GABP	wgEncodeAwgTfbsHaibA549GabpV0422111Etoh02UniPk	18315	4579
	wgEncodeAwgTfbsHaibGm12878GabpPcr2xUniPk	10211	2553
	wgEncodeAwgTfbsHaibH1hesGabpPcr1xUniPk	8968	2242
	wgEncodeAwgTfbsHaibHelas3GabpPcr1xUniPk	10494	2624
	wgEncodeAwgTfbsHaibHepg2GabpPcr2xUniPk	15710	3928
	wgEncodeAwgTfbsHaibK562GabpV0416101UniPk	22387	5597
GATA2	wgEncodeAwgTfbsHaibK562Gata2sc267Pcr1xUniPk	30081	7521
	wgEncodeAwgTfbsSydhHuvecGata2UcdUniPk	42681	10671
	wgEncodeAwgTfbsSydhK562Gata2UcdUniPk	16649	4163
	wgEncodeAwgTfbsSydhShsy5yGata2UcdUniPk	55936	13984
	wgEncodeAwgTfbsUchicagoK562Egata2UniPk	18004	4502
GATA3	wgEncodeAwgTfbsHaibT47dGata3sc268V0416102Dm002p1hUniPk	58292	14574
	wgEncodeAwgTfbsSydhMcf7Gata3UcdUniPk	9585	2397
	wgEncodeAwgTfbsSydhShsy5yGata3sc269sc269UcdUniPk	24948	6238
HAE2F1	wgEncodeAwgTfbsSydhHelas3Hae2f1UniPk	15990	3998
	wgEncodeAwgTfbsSydhMcf7Hae2f1UcdUniPk	30704	7676
HDAC	wgEncodeAwgTfbsBroadK562Hdac2a300705aUniPk	8380	2096
	wgEncodeAwgTfbsBroadK562Hdac6a301341aUniPk	1779	445
	wgEncodeAwgTfbsHaibH1hesHdac2sc6296V0416102UniPk	8953	2239
	wgEncodeAwgTfbsHaibHepg2Hdac2sc6296V0416101UniPk	29865	7467
	wgEncodeAwgTfbsUchicagoK562Ehdac8UniPk	2734	684
	wgEncodeAwgTfbsHaibK562Hdac2sc6296V0416102UniPk	10539	2635
CJUN	wgEncodeAwgTfbsSydhHelas3CjunIggrabUniPk	34089	8523
	wgEncodeAwgTfbsSydhHepg2CjunIggrabUniPk	20036	5010
	wgEncodeAwgTfbsSydhHuvecCjunUniPk	46128	11532
	wgEncodeAwgTfbsSydhK562CjunUniPk	15460	3866
JUND	wgEncodeAwgTfbsHaibHepg2JundPcr1xUniPk	34142	8536
	wgEncodeAwgTfbsSydhGm12878JundUniPk	3892	974
	wgEncodeAwgTfbsSydhH1hesJundIggrabUniPk	15019	3755
	wgEncodeAwgTfbsSydhHelas3JundIggrabUniPk	49201	12301
	wgEncodeAwgTfbsSydhHepg2JundIggrabUniPk	50878	12720
	wgEncodeAwgTfbsSydhK562JundIggrabUniPk	62753	15689
	wgEncodeAwgTfbsUchicagoK562EjundUniPk	41771	10443
MAX	wgEncodeAwgTfbsHaibK562MaxV0416102UniPk	70233	17559

	wgEncodeAwgTfbsSydhGm12878MaxIggmusUniPk	19747	4937
	wgEncodeAwgTfbsSydhH1hescMaxUcdUniPk	17553	4389
	wgEncodeAwgTfbsSydhHelas3MaxIggrabUniPk	46209	11553
	wgEncodeAwgTfbsSydhHepg2MaxIggrabUniPk	18774	4694
	wgEncodeAwgTfbsSydhHuvecMaxUniPk	14409	3603
	wgEncodeAwgTfbsSydhK562MaxIggrabUniPk	49299	12325
	wgEncodeAwgTfbsSydhNb4MaxUniPk	54438	13610
CMYC	wgEncodeAwgTfbsSydhNb4CmycUniPk	41251	10313
	wgEncodeAwgTfbsUtaGm12878CmycUniPk	5830	1458
	wgEncodeAwgTfbsUtaH1hescCmycUniPk	1939	485
	wgEncodeAwgTfbsUtaHepg2CmycUniPk	7012	1754
NFKB	wgEncodeAwgTfbsSydhGm10847NfkbTnfalIggrabUniPk	14545	3637
	wgEncodeAwgTfbsSydhGm12878NfkbTnfalIggrabUniPk	26700	6676
	wgEncodeAwgTfbsSydhGm12891NfkbTnfalIggrabUniPk	45032	11258
	wgEncodeAwgTfbsSydhGm12892NfkbTnfalIggrabUniPk	12865	3217
	wgEncodeAwgTfbsSydhGm18526NfkbTnfalIggrabUniPk	4820	1206
	wgEncodeAwgTfbsSydhGm19099NfkbTnfalIggrabUniPk	12190	3048
PAX5	wgEncodeAwgTfbsHaibGm12878Pax5c20Pcr1xUniPk	39841	9961
	wgEncodeAwgTfbsHaibGm12878Pax5n19Pcr1xUniPk	31012	7754
	wgEncodeAwgTfbsHaibGm12891Pax5c20V0416101UniPk	4979	1245
	wgEncodeAwgTfbsHaibGm12892Pax5c20V0416101UniPk	16284	4072
POL2B	wgEncodeAwgTfbsBroadHelas3Pol2bUniPk	4779	1195
	wgEncodeAwgTfbsBroadHuvecPol2bUniPk	8334	2084
	wgEncodeAwgTfbsBroadNhekPol2bUniPk	9046	2262
P300	wgEncodeAwgTfbsHaibGm12878P300Pcr1xUniPk	8052	2014
	wgEncodeAwgTfbsHaibHepg2P300V0416101UniPk	43721	10931
	wgEncodeAwgTfbsHaibSknshraP300V0416102UniPk	73086	18272
	wgEncodeAwgTfbsHaibT47dP300V0416102Dm002p1hUniPk	21969	5493
	wgEncodeAwgTfbsSydhGm12878P300bUniPk	9697	2425
	wgEncodeAwgTfbsSydhHelas3P300sc584sc584IggrabUniPk	40262	10066
	wgEncodeAwgTfbsSydhK562P300IggrabUniPk	40425	10107
SIN3A	wgEncodeAwgTfbsHaibA549Sin3ak20V0422111Etoh02UniPk	9516	2380
	wgEncodeAwgTfbsHaibHepg2Sin3ak20Pcr1xUniPk	26051	6513
	wgEncodeAwgTfbsHaibK562Sin3ak20V0416101UniPk	19993	4999
	wgEncodeAwgTfbsHaibPanc1Sin3ak20V0416101UniPk	11289	2823
	wgEncodeAwgTfbsSydhH1hescSin3anb6001263IggrabUniPk	33697	8425
SP1	wgEncodeAwgTfbsHaibH1hescSp1Pcr1xUniPk	23648	5912
	wgEncodeAwgTfbsHaibHepg2Sp1Pcr1xUniPk	39952	9988
SP2	wgEncodeAwgTfbsHaibH1hescSp2V0422111UniPk	3830	958
	wgEncodeAwgTfbsHaibHepg2Sp2V0422111UniPk	4152	1038
SRF	wgEncodeAwgTfbsHaibH1hescSrfPcr1xUniPk	8009	2003
	wgEncodeAwgTfbsHaibHepg2SrfV0416101UniPk	8372	2094
	wgEncodeAwgTfbsHaibK562SrfV0416101UniPk	7414	1854
STAT3	wgEncodeAwgTfbsSydhGm12878Stat3IggmusUniPk	10185	2547
	wgEncodeAwgTfbsSydhHelas3Stat3IggrabUniPk	21563	5391
	wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01bUniPk	70411	17603
	wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01cUniPk	64587	16147
	wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01UniPk	19377	4845

	wgEncodeAwgTfbsSydhMcf10aesStat3Tam112hHvdUniPk	62265	15567
	wgEncodeAwgTfbsSydhMcf10aesStat3TamUniPk	69246	17312
TAF1	wgEncodeAwgTfbsHaibA549Taf1V0422111Etoh02UniPk	15691	3923
	wgEncodeAwgTfbsHaibGm12878Taf1Pcr1xUniPk	22305	5577
	wgEncodeAwgTfbsHaibGm12892Taf1V0416102UniPk	14776	3694
	wgEncodeAwgTfbsHaibHepg2Taf1Pcr2xUniPk	25369	6343
	wgEncodeAwgTfbsHaibK562Taf1V0416101UniPk	23428	5858
	wgEncodeAwgTfbsHaibSknsHaf1V0416101UniPk	21908	5478
TBP	wgEncodeAwgTfbsSydhGm12878TbpIggmusUniPk	23278	5820
	wgEncodeAwgTfbsSydhHepg2TbpIggrabUniPk	21732	5434
	wgEncodeAwgTfbsSydhK562TbpIggmusUniPk	27345	6837
	wgEncodeAwgTfbsSydhH1hescCtbp2UcdUniPk	11228	2808
TEAD4	wgEncodeAwgTfbsHaibH1hescTead4sc101184V0422111UniPk	31233	7809
	wgEncodeAwgTfbsHaibHepg2Tead4sc101184V0422111UniPk	23852	5964
	wgEncodeAwgTfbsHaibK562Tead4sc101184V0422111UniPk	48372	12094
YY1	wgEncodeAwgTfbsHaibGm12878Yy1sc281Pcr1xUniPk	48142	12036
	wgEncodeAwgTfbsHaibGm12892Yy1V0416101UniPk	24763	6191
	wgEncodeAwgTfbsHaibH1hescYy1sc281V0416102UniPk	28672	7168
	wgEncodeAwgTfbsHaibHct116Yy1sc281V0416101UniPk	20043	5011
	wgEncodeAwgTfbsHaibHepg2Yy1sc281V0416101UniPk	27944	6986
	wgEncodeAwgTfbsHaibK562Yy1V0416101UniPk	19684	4922
	wgEncodeAwgTfbsHaibK562Yy1V0416102UniPk	37296	9324
	wgEncodeAwgTfbsHaibSknsHraYy1sc281V0416102UniPk	24422	6106
	wgEncodeAwgTfbsSydhK562Yy1UcdUniPk	7638	1910
	wgEncodeAwgTfbsSydhNt2d1Yy1UcdUniPk	7465	1867

Table S2: Positive and negative sample distributions of benchmark datasets

TFs	Motif_dataset	Train_dataset		Test_dataset	
		Number of positive samples	Number of negative samples	Number of positive samples	Number of negative samples
CEBPB	wgEncodeAwgTfbsHaibHepg2CebpbSc150V0416101UniPk	14243	14248	3564	3559
	wgEncodeAwgTfbsHaibK562CebpbSc150V0422111UniPk	17353	17394	4364	4323
	wgEncodeAwgTfbsSydhA549CebpbIggrabUniPk	30409	30336	7557	7630
	wgEncodeAwgTfbsSydhH1hesCebpbIggrabUniPk	12295	12177	3001	3117
	wgEncodeAwgTfbsSydhHelas3CebpbIggrabUniPk	47379	47513	11930	11794
	wgEncodeAwgTfbsSydhHepg2CebpbForsklnUniPk	14496	14516	3638	3616
	wgEncodeAwgTfbsSydhHepg2CebpbIggrabUniPk	44654	44461	11044	11235
	wgEncodeAwgTfbsSydhImr90CebpbIggrabUniPk	54861	54851	13710	13718
	wgEncodeAwgTfbsSydhK562CebpbIggrabUniPk	30270	30146	7492	7612
COREST	wgEncodeAwgTfbsSydhHepg2Corestsc30189IggrabUniPk	4271	4244	1051	1078
	wgEncodeAwgTfbsSydhK562Corestsc30189IggrabUniPk	28109	28102	7023	7030
CTCF	wgEncodeAwgTfbsBroadDnd41CtcfUniPk	38783	38748	9674	9709
	wgEncodeAwgTfbsBroadGm12878CtcfUniPk	34753	34682	8644	8715
	wgEncodeAwgTfbsBroadH1hesCtcfUniPk	51646	51528	12838	12956
	wgEncodeAwgTfbsBroadHelas3CtcfUniPk	40493	40556	10163	10100
	wgEncodeAwgTfbsBroadHepg2CtcfUniPk	36196	36128	9007	9075
	wgEncodeAwgTfbsBroadHmecCtcfUniPk	30852	30949	7774	7677
	wgEncodeAwgTfbsBroadHsmtCtcfUniPk	39592	39444	9806	9954
	wgEncodeAwgTfbsBroadHsmmtCtcfUniPk	37327	37077	9176	9426
	wgEncodeAwgTfbsBroadHuvecCtcfUniPk	29058	29044	7256	7270
	wgEncodeAwgTfbsBroadK562CtcfUniPk	40042	40375	10219	9886
	wgEncodeAwgTfbsBroadNhaCtcfUniPk	29817	29835	7466	7448
	wgEncodeAwgTfbsBroadNhfdCtcfUniPk	37984	37867	9423	9540
	wgEncodeAwgTfbsBroadNhekCtcfUniPk	36566	36389	9031	9208
	wgEncodeAwgTfbsBroadNhlfCtcfUniPk	30645	30692	7691	7644
	wgEncodeAwgTfbsBroadOsteoblCtcfUniPk	42935	43213	10908	10630
	wgEncodeAwgTfbsUtaGm12878CtcfUniPk	37790	37746	9420	9464
	wgEncodeAwgTfbsUtaH1hesCtcfUniPk	33756	33845	8495	8406
	wgEncodeAwgTfbsUtaHelas3CtcfUniPk	45460	45423	11343	11378
	wgEncodeAwgTfbsUtaK562CtcfUniPk	43374	43434	10881	10821
	wgEncodeAwgTfbsUtaMcf7CtcfUniPk	45057	44992	11224	11289
ELF1	wgEncodeAwgTfbsHaibA549Elf1V0422111Etoh02UniPk	6782	6686	1636	1732
	wgEncodeAwgTfbsHaibGm12878Elf1sc631V0416101UniPk	17857	17827	4446	4476
	wgEncodeAwgTfbsHaibHepg2Elf1sc631V0416101UniPk	14232	14083	3465	3614

	wgEncodeAwgTfbsHaibK562Elf1sc631V0416102UniPk	21787	21616	5340	5511
EZH2	wgEncodeAwgTfbsBroadDnd41Ezh239875UniPk	1329	1352	348	323
	wgEncodeAwgTfbsBroadHelas3Ezh239875UniPk	1404	1436	371	339
	wgEncodeAwgTfbsBroadHepg2Ezh239875UniPk	2618	2617	654	655
	wgEncodeAwgTfbsBroadHsmmEzh239875UniPk	1239	1225	300	316
	wgEncodeAwgTfbsBroadHuvecEzh239875UniPk	3397	3391	845	853
	wgEncodeAwgTfbsBroadK562Ezh239875UniPk	1366	1318	311	361
FOS	wgEncodeAwgTfbsHaibA549Fosl2V0422111Etoh02UniPk	22355	22395	5614	5574
	wgEncodeAwgTfbsHaibH1hesCFosl1sc183V0416102UniPk	875	883	224	216
	wgEncodeAwgTfbsHaibHepg2Fosl2V0416101UniPk	19899	19779	4900	5020
	wgEncodeAwgTfbsHaibK562Fosl1sc183V0416101UniPk	8782	8787	2199	2194
	wgEncodeAwgTfbsSydhGm12878CfosUniPk	1749	1739	431	441
	wgEncodeAwgTfbsSydhHelas3CfosUniPk	7251	7243	1808	1816
	wgEncodeAwgTfbsSydhHuvecCfosUcdUniPk	36461	36500	9140	9101
	wgEncodeAwgTfbsSydhK562CfosUniPk	5985	5983	1495	1497
	wgEncodeAwgTfbsSydhMcf10aesCfosTam112hHvdUniPk	71850	71575	17791	18066
	wgEncodeAwgTfbsSydhMcf10aesCfosTam14hHvdUniPk	67066	67014	16734	16786
	wgEncodeAwgTfbsSydhMcf10aesCfosTamHvdUniPk	54707	54747	13702	13662
	wgEncodeAwgTfbsUchicagoK562EfosUniPk	8064	8056	2011	2019
FOXA1	wgEncodeAwgTfbsHaibA549Foxa1V0416102Dex100nmUniPk	6009	6045	1525	1489
	wgEncodeAwgTfbsHaibEcc1Foxa1sc6553V0416102Dm002p1hUniPk	4671	4612	1131	1190
	wgEncodeAwgTfbsHaibHepg2Foxa1sc101058V0416101UniPk	33729	33821	8490	8398
	wgEncodeAwgTfbsHaibHepg2Foxa1sc6553V0416101UniPk	40132	40092	10008	10048
	wgEncodeAwgTfbsHaibT47dFoxa1sc6553V0416102Dm002p1hUniPk	32624	32633	8162	8153
GABP	wgEncodeAwgTfbsHaibA549GabpV0422111Etoh02UniPk	9203	9112	2244	2335
	wgEncodeAwgTfbsHaibGm12878GabpPcr2xUniPk	5106	5105	1276	1277
	wgEncodeAwgTfbsHaibH1hesCGabpPcr1xUniPk	4490	4478	1115	1127
	wgEncodeAwgTfbsHaibHelas3GabpPcr1xUniPk	5231	5263	1328	1296
	wgEncodeAwgTfbsHaibHepg2GabpPcr2xUniPk	7871	7839	1948	1980
	wgEncodeAwgTfbsHaibK562GabpV0416101UniPk	11191	11196	2801	2796
GATA2	wgEncodeAwgTfbsHaibK562Gata2sc267Pcr1xUniPk	15080	15001	3721	3800
	wgEncodeAwgTfbsSydhHuvecGata2UcdUniPk	21339	21342	5337	5334
	wgEncodeAwgTfbsSydhK562Gata2UcdUniPk	8323	8326	2083	2080
	wgEncodeAwgTfbsSydhShsy5yGata2UcdUniPk	27935	28001	7025	6959
	wgEncodeAwgTfbsUchicagoK562Egata2UniPk	9008	8996	2245	2257
GATA3	wgEncodeAwgTfbsHaibT47dGata3sc268V0416102Dm002p1hUniPk	29070	29222	7363	7211
	wgEncodeAwgTfbsSydhMcf7Gata3UcdUniPk	4793	4792	1198	1199
	wgEncodeAwgTfbsSydhShsy5yGata3sc269sc269UcdUniPk	12440	12508	3153	3085

HAE2F1	wgEncodeAwgTfbsSydhHelas3Hae2f1UniPk	7993	7997	2001	1997
	wgEncodeAwgTfbsSydhMcf7Hae2f1UcdUniPk	15377	15327	3813	3863
HDAC	wgEncodeAwgTfbsBroadK562Hdac2a300705aUniPk	4185	4195	1053	1043
	wgEncodeAwgTfbsBroadK562Hdac6a301341aUniPk	884	895	228	217
	wgEncodeAwgTfbsHaibH1hescHdac2sc6296V0416102UniPk	4472	4481	1124	1115
	wgEncodeAwgTfbsHaibHepg2Hdac2sc6296V0416101UniPk	14926	14939	3740	3727
	wgEncodeAwgTfbsUchicagoK562Ehdac8UniPk	1376	1358	333	351
	wgEncodeAwgTfbsHaibK562Hdac2sc6296V0416102UniPk	5288	5251	1299	1336
JUN	wgEncodeAwgTfbsSydhHelas3CjunIggrabUniPk	17057	17032	4249	4274
	wgEncodeAwgTfbsSydhHepg2CjunIggrabUniPk	10001	10035	2522	2488
	wgEncodeAwgTfbsSydhHuvecCjunUniPk	23077	23051	5753	5779
	wgEncodeAwgTfbsSydhK562CjunUniPk	7788	7672	1875	1991
JUND	wgEncodeAwgTfbsHaibHepg2JundPcr1xUniPk	17095	17047	4244	4292
	wgEncodeAwgTfbsSydhGm12878JundUniPk	1974	1918	459	515
	wgEncodeAwgTfbsSydhH1hescJundIggrabUniPk	7532	7487	1855	1900
	wgEncodeAwgTfbsSydhHelas3JundIggrabUniPk	24647	24554	6104	6197
	wgEncodeAwgTfbsSydhHepg2JundIggrabUniPk	25448	25430	6351	6369
	wgEncodeAwgTfbsSydhK562JundIggrabUniPk	31391	31362	7830	7859
	wgEncodeAwgTfbsUchicagoK562EjundUniPk	20954	20817	5153	5290
MAX	wgEncodeAwgTfbsHaibK562MaxV0416102UniPk	34919	35314	8977	8582
	wgEncodeAwgTfbsSydhGm12878MaxIggmusUniPk	9872	9875	2470	2467
	wgEncodeAwgTfbsSydhH1hescMaxUcdUniPk	8709	8844	2262	2127
	wgEncodeAwgTfbsSydhHelas3MaxIggrabUniPk	23129	23080	5752	5801
	wgEncodeAwgTfbsSydhHepg2MaxIggrabUniPk	9382	9392	2352	2342
	wgEncodeAwgTfbsSydhHuvecMaxUniPk	7214	7195	1792	1811
	wgEncodeAwgTfbsSydhK562MaxIggrabUniPk	24653	24646	6159	6166
	wgEncodeAwgTfbsSydhNb4MaxUniPk	27209	27229	6815	6795
MYC	wgEncodeAwgTfbsSydhNb4CmycUniPk	20572	20679	5210	5103
	wgEncodeAwgTfbsUtaGm12878CmycUniPk	2923	2907	721	737
	wgEncodeAwgTfbsUtaH1hescCmycUniPk	971	968	241	244
	wgEncodeAwgTfbsUtaHepg2CmycUniPk	3509	3503	874	880
NFKB	wgEncodeAwgTfbsSydhGm10847NfkbTnfalIggrabUniPk	7234	7311	1857	1780
	wgEncodeAwgTfbsSydhGm12878NfkbTnfalIggrabUniPk	13345	13355	3343	3333
	wgEncodeAwgTfbsSydhGm12891NfkbTnfalIggrabUniPk	22489	22543	5656	5602
	wgEncodeAwgTfbsSydhGm12892NfkbTnfalIggrabUniPk	6423	6442	1618	1599
	wgEncodeAwgTfbsSydhGm18526NfkbTnfalIggrabUniPk	2395	2425	617	589
	wgEncodeAwgTfbsSydhGm19099NfkbTnfalIggrabUniPk	6073	6117	1546	1502
Pax5	wgEncodeAwgTfbsHaibGm12878Pax5c20Pcr1xUniPk	20004	19837	4897	5064
	wgEncodeAwgTfbsHaibGm12878Pax5n19Pcr1xUniPk	15487	15525	3896	3858
	wgEncodeAwgTfbsHaibGm12891Pax5c20V0416101UniPk	2497	2482	615	630

	wgEncodeAwgTfbsHaibGm12892Pax5c20V0416101UniPk	8076	8208	2102	1970
POL2B	wgEncodeAwgTfbsBroadHelas3Pol2bUniPk	2385	2394	602	593
	wgEncodeAwgTfbsBroadHuvecPol2bUniPk	4187	4147	1022	1062
	wgEncodeAwgTfbsBroadNhekPol2bUniPk	4516	4530	1138	1124
P300	wgEncodeAwgTfbsHaibGm12878P300Pcr1xUniPk	4034	4018	999	1015
	wgEncodeAwgTfbsHaibHepg2P300V0416101UniPk	21887	21834	5439	5492
	wgEncodeAwgTfbsHaibSknshraP300V0416102UniPk	36487	36599	9192	9080
	wgEncodeAwgTfbsHaibT47dP300V0416102Dm002p1hUniPk	10977	10992	2754	2739
	wgEncodeAwgTfbsSydhGm12878P300bUniPk	4880	4817	1181	1244
	wgEncodeAwgTfbsSydhHelas3P300sc584sc584IggrabUniPk	20156	20106	5008	5058
	wgEncodeAwgTfbsSydhK562P300IggrabUniPk	20236	20189	5030	5077
SIN3A	wgEncodeAwgTfbsHaibA549Sin3ak20V0422111Etoh02UniPk	4765	4751	1183	1197
	wgEncodeAwgTfbsHaibHepg2Sin3ak20Pcr1xUniPk	13007	13044	3275	3238
	wgEncodeAwgTfbsHaibK562Sin3ak20V0416101UniPk	9982	10011	2514	2485
	wgEncodeAwgTfbsHaibPanc1Sin3ak20V0416101UniPk	5690	5599	1366	1457
	wgEncodeAwgTfbsSydhH1hesccSin3anb6001263IggrabUniPk	16812	16885	4249	4176
SP1	wgEncodeAwgTfbsHaibH1hesccSp1Pcr1xUniPk	11833	11815	2947	2965
	wgEncodeAwgTfbsHaibHepg2Sp1Pcr1xUniPk	19891	20061	5079	4909
SP2	wgEncodeAwgTfbsHaibH1hesccSp2V0422111UniPk	1915	1915	479	479
	wgEncodeAwgTfbsHaibHepg2Sp2V0422111UniPk	2085	2067	510	528
SRF	wgEncodeAwgTfbsHaibH1hesccSrfPcr1xUniPk	4008	4001	998	1005
	wgEncodeAwgTfbsHaibHepg2SrfV0416101UniPk	4203	4169	1030	1064
	wgEncodeAwgTfbsHaibK562SrfV0416101UniPk	3725	3689	909	945
STAT3	wgEncodeAwgTfbsSydhGm12878Stat3IggmusUniPk	5118	5067	1248	1299
	wgEncodeAwgTfbsSydhHelas3Stat3IggrabUniPk	10740	10823	2737	2654
	wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01bUniPk	35281	35130	8726	8877
	wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01cUniPk	32324	32263	8043	8104
	wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01UniPk	9713	9664	2398	2447
	wgEncodeAwgTfbsSydhMcf10aesStat3Tam112hHvdUniPk	31109	31156	7807	7760
	wgEncodeAwgTfbsSydhMcf10aesStat3TamUniPk	34662	34584	8617	8695
TAF1	wgEncodeAwgTfbsHaibA549Taf1V0422111Etoh02UniPk	7833	7858	1974	1949
	wgEncodeAwgTfbsHaibGm12878Taf1Pcr1xUniPk	11154	11151	2787	2790
	wgEncodeAwgTfbsHaibGm12892Taf1V0416102UniPk	7381	7395	1854	1840
	wgEncodeAwgTfbsHaibHepg2Taf1Pcr2xUniPk	12702	12667	3154	3189
	wgEncodeAwgTfbsHaibK562Taf1V0416101UniPk	11726	11702	2917	2941
	wgEncodeAwgTfbsHaibSknshTaf1V0416101UniPk	11002	10906	2691	2787
TBPI	wgEncodeAwgTfbsSydhGm12878TbpIggmusUniPk	11684	11594	2865	2955
	wgEncodeAwgTfbsSydhHepg2TbpIggrabUniPk	10857	10875	2726	2708
	wgEncodeAwgTfbsSydhK562TbpIggmusUniPk	13654	13691	3437	3400
	wgEncodeAwgTfbsSydhH1hesccTbp2UcdUniPk	5625	5603	1393	1415

TEAD4	wgEncodeAwgTfbsHaibH1hescTead4sc101184V0422111UniPk	15632	15601	3889	3920
	wgEncodeAwgTfbsHaibHepg2Tead4sc101184V0422111UniPk	11933	11919	2975	2989
	wgEncodeAwgTfbsHaibK562Tead4sc101184V0422111UniPk	24181	24191	6052	6042
YY1	wgEncodeAwgTfbsHaibGm12878Yy1sc281Pcr1xUniPk	24010	24132	6079	5957
	wgEncodeAwgTfbsHaibGm12892Yy1V0416101UniPk	12306	12457	3171	3020
	wgEncodeAwgTfbsHaibH1hescYy1sc281V0416102UniPk	14357	14315	3563	3605
	wgEncodeAwgTfbsHaibHct116Yy1sc281V0416101UniPk	9998	10045	2529	2482
	wgEncodeAwgTfbsHaibHepg2Yy1sc281V0416101UniPk	14065	13879	3400	3586
	wgEncodeAwgTfbsHaibK562Yy1V0416101UniPk	9882	9802	2421	2501
	wgEncodeAwgTfbsHaibK562Yy1V0416102UniPk	18598	18698	4712	4612
	wgEncodeAwgTfbsHaibSknshraYy1sc281V0416102UniPk	12189	12233	3075	3031
	wgEncodeAwgTfbsSydhK562Yy1UcdUniPk	3812	3826	962	948
	wgEncodeAwgTfbsSydhNt2d1Yy1UcdUniPk	3690	3775	976	891

Table S3: 165 ChIP-seq datasets belong to cell lines and source organism information

Cell lines	Datasets	Source organism
HepG2	wgEncodeAwgTfbsBroadHepg2CtcfUniPk	Human Hepatocellular Carcinoma Cells
	wgEncodeAwgTfbsBroadHepg2Ezh239875UniPk	
	wgEncodeAwgTfbsHaibHepg2Cebpbsc150V0416101UniPk	
	wgEncodeAwgTfbsHaibHepg2Elf1sc631V0416101UniPk	
	wgEncodeAwgTfbsHaibHepg2Fosl2V0416101UniPk	
	wgEncodeAwgTfbsHaibHepg2Foxa1sc101058V0416101UniPk	
	wgEncodeAwgTfbsHaibHepg2Foxa1sc6553V0416101UniPk	
	wgEncodeAwgTfbsHaibHepg2GabpPcr2xUniPk	
	wgEncodeAwgTfbsHaibHepg2Hdac2sc6296V0416101UniPk	
	wgEncodeAwgTfbsHaibHepg2JundPcr1xUniPk	
	wgEncodeAwgTfbsHaibHepg2P300V0416101UniPk	
	wgEncodeAwgTfbsHaibHepg2Sin3ak20Pcr1xUniPk	
	wgEncodeAwgTfbsHaibHepg2Sp1Pcr1xUniPk	
	wgEncodeAwgTfbsHaibHepg2Sp2V0422111UniPk	
	wgEncodeAwgTfbsHaibHepg2SrfV0416101UniPk	
	wgEncodeAwgTfbsHaibHepg2Taf1Pcr2xUniPk	
	wgEncodeAwgTfbsHaibHepg2Tead4sc101184V0422111UniPk	
	wgEncodeAwgTfbsHaibHepg2Yy1sc281V0416101UniPk	
	wgEncodeAwgTfbsSydhHepg2CebpbforsklnUniPk	
	wgEncodeAwgTfbsSydhHepg2CebpblggrabUniPk	
	wgEncodeAwgTfbsSydhHepg2CjunlggrabUniPk	
	wgEncodeAwgTfbsSydhHepg2Corestsc30189lggrabUniPk	
	wgEncodeAwgTfbsSydhHepg2JundlggrabUniPk	
	wgEncodeAwgTfbsSydhHepg2MaxlggrabUniPk	
	wgEncodeAwgTfbsSydhHepg2TbplggrabUniPk	
	wgEncodeAwgTfbsUtaHepg2CmycUniPk	
K562	wgEncodeAwgTfbsBroadK562CtcfUniPk	Human Chronic Myelogenous Leukemia Cells
	wgEncodeAwgTfbsBroadK562Ezh239875UniPk	
	wgEncodeAwgTfbsBroadK562Hdac2a300705aUniPk	
	wgEncodeAwgTfbsBroadK562Hdac6a301341aUniPk	
	wgEncodeAwgTfbsHaibK562Cebpbsc150V0422111UniPk	
	wgEncodeAwgTfbsHaibK562Elf1sc631V0416102UniPk	
	wgEncodeAwgTfbsHaibK562Fosl1sc183V0416101UniPk	
	wgEncodeAwgTfbsHaibK562GabpV0416101UniPk	
	wgEncodeAwgTfbsHaibK562Gata2sc267Pcr1xUniPk	
	wgEncodeAwgTfbsHaibK562Hdac2sc6296V0416102UniPk	
	wgEncodeAwgTfbsHaibK562MaxV0416102UniPk	
	wgEncodeAwgTfbsHaibK562Sin3ak20V0416101UniPk	
	wgEncodeAwgTfbsHaibK562SrfV0416101UniPk	
	wgEncodeAwgTfbsHaibK562Taf1V0416101UniPk	
	wgEncodeAwgTfbsHaibK562Tead4sc101184V0422111UniPk	
	wgEncodeAwgTfbsHaibK562Yy1V0416101UniPk	
	wgEncodeAwgTfbsHaibK562Yy1V0416102UniPk	
	wgEncodeAwgTfbsSydhK562CebpblggrabUniPk	
	wgEncodeAwgTfbsSydhK562CfosUniPk	

	wgEncodeAwgTfbsSydhK562CjunUniPk	
	wgEncodeAwgTfbsSydhK562Corestsc30189IggrabUniPk	
	wgEncodeAwgTfbsSydhK562Gata2UcdUniPk	
	wgEncodeAwgTfbsSydhK562JundIggrabUniPk	
	wgEncodeAwgTfbsSydhK562MaxIggrabUniPk	
	wgEncodeAwgTfbsSydhK562P300IggrabUniPk	
	wgEncodeAwgTfbsSydhK562TbpIggmusUniPk	
	wgEncodeAwgTfbsSydhK562Yy1UcdUniPk	
	wgEncodeAwgTfbsUchicagoK562EfosUniPk	
	wgEncodeAwgTfbsUchicagoK562Egata2UniPk	
	wgEncodeAwgTfbsUchicagoK562Ehdac8UniPk	
	wgEncodeAwgTfbsUchicagoK562EjundUniPk	
	wgEncodeAwgTfbsUtaK562CtcfUniPk	
A549	wgEncodeAwgTfbsHaibA549Elf1V0422111EtOH02UniPk	Human Lung Adenocarcinoma Cells
	wgEncodeAwgTfbsHaibA549Fosl2V0422111EtOH02UniPk	
	wgEncodeAwgTfbsHaibA549Foxa1V0416102Dex100nmUniPk	
	wgEncodeAwgTfbsHaibA549GabpV0422111EtOH02UniPk	
	wgEncodeAwgTfbsHaibA549Sin3ak20V0422111EtOH02UniPk	
	wgEncodeAwgTfbsHaibA549Taf1V0422111EtOH02UniPk	
H1hesc	wgEncodeAwgTfbsSydhA549CebpIggrabUniPk	Human Embryonic Stem Cells
	wgEncodeAwgTfbsBroadH1hescCtcfUniPk	
	wgEncodeAwgTfbsHaibH1hescFosl1sc183V0416102UniPk	
	wgEncodeAwgTfbsHaibH1hescGabpPcr1xUniPk	
	wgEncodeAwgTfbsHaibH1hescHdac2sc6296V0416102UniPk	
	wgEncodeAwgTfbsHaibH1hescSp1Pcr1xUniPk	
	wgEncodeAwgTfbsHaibH1hescSp2V0422111UniPk	
	wgEncodeAwgTfbsHaibH1hescSrfPcr1xUniPk	
	wgEncodeAwgTfbsHaibH1hescTead4sc101184V0422111UniPk	
	wgEncodeAwgTfbsHaibH1hescYy1sc281V0416102UniPk	
	wgEncodeAwgTfbsSydhH1hescCebpIggrabUniPk	
	wgEncodeAwgTfbsSydhH1hescCtbp2UcdUniPk	
	wgEncodeAwgTfbsSydhH1hescJundIggrabUniPk	
	wgEncodeAwgTfbsSydhH1hescMaxUcdUniPk	
HelaS3	wgEncodeAwgTfbsSydhH1hescSin3anb6001263IggrabUniPk	Human Cervical Carcinoma Cells
	wgEncodeAwgTfbsUtaH1hescCmycUniPk	
	wgEncodeAwgTfbsUtaH1hescCtcfUniPk	
	wgEncodeAwgTfbsBroadHelaS3CtcfUniPk	
	wgEncodeAwgTfbsBroadHelaS3Ezh239875UniPk	
	wgEncodeAwgTfbsBroadHelaS3Pol2bUniPk	
	wgEncodeAwgTfbsHaibHelaS3GabpPcr1xUniPk	
	wgEncodeAwgTfbsSydhHelaS3CebpIggrabUniPk	
	wgEncodeAwgTfbsSydhHelaS3CfosUniPk	
	wgEncodeAwgTfbsSydhHelaS3CjunIggrabUniPk	
	wgEncodeAwgTfbsSydhHelaS3Hae2f1UniPk	
	wgEncodeAwgTfbsSydhHelaS3JundIggrabUniPk	
	wgEncodeAwgTfbsSydhHelaS3MaxIggrabUniPk	
	wgEncodeAwgTfbsSydhHelaS3P300sc584sc584IggrabUniPk	

	wgEncodeAwgTfbsSydhHelas3Stat3IggrabUniPk	
	wgEncodeAwgTfbsUtaHelas3CtcfUniPk	
IMR90	wgEncodeAwgTfbsSydhImr90CebpbIggrabUniPk	Human Fetal Lung Fibroblast Cells
Dnd41	wgEncodeAwgTfbsBroadDnd41CtcfUniPk	
	wgEncodeAwgTfbsBroadDnd41Ezh239875UniPk	Human T-cell Leukemia Cells
	wgEncodeAwgTfbsBroadGm12878CtcfUniPk	
	wgEncodeAwgTfbsHaibGm12878Elf1sc631V0416101UniPk	
	wgEncodeAwgTfbsHaibGm12878GabpPcr2xUniPk	
	wgEncodeAwgTfbsHaibGm12878P300Pcr1xUniPk	
	wgEncodeAwgTfbsHaibGm12878Pax5c20Pcr1xUniPk	
	wgEncodeAwgTfbsHaibGm12878Pax5n19Pcr1xUniPk	
	wgEncodeAwgTfbsHaibGm12878Taf1Pcr1xUniPk	
	wgEncodeAwgTfbsHaibGm12878Yy1sc281Pcr1xUniPk	
Gm12878	wgEncodeAwgTfbsSydhGm12878CfosUniPk	Human B-Lymphoblastoid Cells
	wgEncodeAwgTfbsSydhGm12878JundUniPk	
	wgEncodeAwgTfbsSydhGm12878MaxIggmusUniPk	
	wgEncodeAwgTfbsSydhGm12878NfkbTnfalgggrabUniPk	
	wgEncodeAwgTfbsSydhGm12878P300bUniPk	
	wgEncodeAwgTfbsSydhGm12878Stat3IggmusUniPk	
	wgEncodeAwgTfbsSydhGm12878TbpIggmusUniPk	
	wgEncodeAwgTfbsUtaGm12878CmycUniPk	
	wgEncodeAwgTfbsUtaGm12878CtcfUniPk	
Hmec	wgEncodeAwgTfbsBroadHmecCtcfUniPk	Human Mammary Epithelial Cells
	wgEncodeAwgTfbsBroadHsmmCtcfUniPk	
Hsmm	wgEncodeAwgTfbsBroadHsmmEzh239875UniPk	Human Skeletal Muscle Myoblast Cells
	wgEncodeAwgTfbsBroadHsmmtCtcfUniPk	
	wgEncodeAwgTfbsBroadHuvecCtcfUniPk	
	wgEncodeAwgTfbsBroadHuvecEzh239875UniPk	
	wgEncodeAwgTfbsBroadHuvecPol2bUniPk	
Huvec	wgEncodeAwgTfbsSydhHuvecCfosUcdUniPk	Human Umbilical Vein Endothelial Cells
	wgEncodeAwgTfbsSydhHuvecCjunUniPk	
	wgEncodeAwgTfbsSydhHuvecGata2UcdUniPk	
	wgEncodeAwgTfbsSydhHuvecMaxUniPk	
Nha	wgEncodeAwgTfbsBroadNhaCtcfUniPk	Human Adrenal Cortex Cells
Nhdfad	wgEncodeAwgTfbsBroadNhdfadCtcfUniPk	Human Dermal Fibroblast Adult
Nhek	wgEncodeAwgTfbsBroadNhekCtcfUniPk	Human Renal Proximal Tubule Epithelial Cells
	wgEncodeAwgTfbsBroadNhekPol2bUniPk	
Nhlf	wgEncodeAwgTfbsBroadNhlfCtcfUniPk	Human Lung Fibroblast
Osteobl	wgEncodeAwgTfbsBroadOsteoblCtcfUniPk	Human Osteoblast
Ecc1	wgEncodeAwgTfbsHaibEcc1Foxa1sc6553V0416102Dm002p1hUniPk	Human Endometrial Carcinoma Cells
Gm12891	wgEncodeAwgTfbsHaibGm12891Pax5c20V0416101UniPk	
	wgEncodeAwgTfbsSydhGm12891NfkbTnfalgggrabUniPk	Human B-Lymphocyte
	wgEncodeAwgTfbsHaibGm12892Pax5c20V0416101UniPk	
Gm12892	wgEncodeAwgTfbsHaibGm12892Taf1V0416102UniPk	
	wgEncodeAwgTfbsHaibGm12892Yy1V0416101UniPk	Human B-Lymphocyte
	wgEncodeAwgTfbsSydhGm12892NfkbTnfalgggrabUniPk	

Hct116	wgEncodeAwgTfbsHaibHct116Yy1sc281V0416101UniPk	Human Colon Carcinoma Cells
Panc1	wgEncodeAwgTfbsHaibPanc1Sin3ak20V0416101UniPk	Human Pancreatic Carcinoma Cells
Sknshra	wgEncodeAwgTfbsHaibSknshraP300V0416102UniPk wgEncodeAwgTfbsHaibSknshraYy1sc281V0416102UniPk	Human Neuroblastoma Cells
T47d	wgEncodeAwgTfbsHaibT47dP300V0416102Dm002p1hUniPk wgEncodeAwgTfbsHaibT47dFoxa1sc6553V0416102Dm002p1hUniPk wgEncodeAwgTfbsHaibT47dGata3sc268V0416102Dm002p1hUniPk	Human Breast Carcinoma Cells
Gm10847	wgEncodeAwgTfbsSydhGm10847NfkbTnfalgggrabUniPk	Human Prostate Carcinoma Cells
Nb4	wgEncodeAwgTfbsSydhNb4CmycUniPk wgEncodeAwgTfbsSydhNb4MaxUniPk	Human Acute Promyelocytic Leukemia Cells
Sknsh	wgEncodeAwgTfbsHaibSknshTaf1V0416101UniPk	Human Neuroblastoma Cells
Gm18526	wgEncodeAwgTfbsSydhGm18526NfkbTnfalgggrabUniPk	Human Lymphoblast Cells
Gm19099	wgEncodeAwgTfbsSydhGm19099NfkbTnfalgggrabUniPk	Human Lymphoblast Cells
Mcf10aes	wgEncodeAwgTfbsSydhMcf10aesCfosTam112hHvdUniPk wgEncodeAwgTfbsSydhMcf10aesCfosTam14hHvdUniPk wgEncodeAwgTfbsSydhMcf10aesCfosTamHvdUniPk wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01bUniPk wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01cUniPk wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01UniPk wgEncodeAwgTfbsSydhMcf10aesStat3Tam112hHvdUniPk wgEncodeAwgTfbsSydhMcf10aesStat3TamUniPk	Human Mammary Epithelial Cells
Nt2d1	wgEncodeAwgTfbsSydhNt2d1Yy1UcdUniPk	Human Embryonic Carcinoma Cells
Shsy5y	wgEncodeAwgTfbsSydhShsy5yGata2UcdUniPk wgEncodeAwgTfbsSydhShsy5yGata3sc269sc269UcdUniPk	Human Neuroblastoma Cells
Mcf7	wgEncodeAwgTfbsSydhMcf7Gata3UcdUniPk wgEncodeAwgTfbsSydhMcf7Hae2f1UcdUniPk wgEncodeAwgTfbsUtaMcf7CtcfUniPk	Human Breast Carcinoma Cells

Table S4. Learning rate sensitivity analysis

The learning rate is one of the key hyperparameters that affects both the convergence speed and final performance of a model. To identify the optimal learning rate, experiments were performed with values of 0.00005, 0.0001, and 0.0002, and the model's performance was evaluated in terms of Accuracy, ROC-AUC, and PR-AUC, as shown in the following table.

From the experimental results, it is evident that a learning rate of 0.0002 leads to relatively poor performance across all metrics, likely due to training instability or failure to converge to an optimal solution. Conversely, with learning rates of 0.00005 and 0.0001, the model achieves improved performance, showing highly consistent results across all evaluation metrics. Specifically, with a learning rate of 0.0001, the model achieves the highest Accuracy, while its ROC-AUC and PR-AUC are slightly lower than those obtained with a learning rate of 0.00005.

As the learning rate decreases, we observe distinct changes in model behavior. While a learning rate of 0.00005 provides a minor advantage in AUC metrics, training at this rate markedly slows convergence, leading to a substantial increase in overall computational time. Considering the minimal performance differences and the substantial difference in training efficiency, we ultimately chose 0.0001 as the model's learning rate. This selection guarantees strong model performance while also maintaining practical training efficiency and managing computational costs.

Learning rate	ACC	ROC-AUC	PR-AUC
0.00005	0.838115	0.912915	0.915325
0.0001	0.838173	0.912690	0.914850
0.0002	0.836998	0.911293	0.913291

Table S5: Hyperparameter calibration results of MSGSNet

Calibration parameters	Search space	Optimal parameter
Max epochs	200, 300, 400	300
Learning rate	5e-5, 1e-4, 2e-4	1e-4
Batch size	64, 128, 256	Adaptive (64 if $N < 20k$; 128 if $20k \leq N < 60k$; 256 if $N \geq 60k$)
Optimizer	Adam, AdamW, SGD	Adam
Weight decay	1e-6, 1e-5, 1e-4	1e-5
Random seed	0, 42, 2025	42

Note: N denotes the number of training samples in each ChIP-seq dataset.

Table S6: Configuration of the Sequence Semantic Branch

Architectures	Settings	Output shape
multi-convolutional layer1	kernel number = 128	(B,128,99)
	kernel size = 3	
	padding = 1	
multi-convolutional layer2	kernel number = 128	(B,128,99)
	kernel size = 7	
	padding = 3	
multi-convolutional layer3	kernel number = 128	(B,128,99)
	kernel size = 11	
	padding = 5	
Concatenation layer	dim = 1	(B,384,99)
BatchNorm1d layer	features=384	(B,384,99)
Dropout layer	dropout = 0.5	(B,384,99)
BiLSTM layer	hidden_size=128	(B,99,256)
	bidirectional=True	
Hierarchical Bi-Attention layer	num_heads=4	(B,99,256)
	local_window=15	
	dropout=0.5	
Global average pooling layer	dim = 1	(B,256)

Note: B represents the batch size in the output tensor shape.

Table S7: Configuration of the Multi-Scale Graph Branch

Architectures	Settings	Output shape
ChebConv layer1	hidden_channels=128 K=3	(B,128)
ChebConv layer2	hidden_channels=128 K=3	(B,128)
Graph Channel Attention layer	channels = 128 reduction=8	(B,128)
Global mean pool layer	-	(B,128)
Pairwise Cross Attention layer	heads=4 dropout = 0.1	(B,384)
MLP layer	ReLU dropout=0.5 num_classes=2	(B,2)

Note: B represents the batch size in the output tensor shape.

Table S8: Configuration of the Cross-Modal Fusion Module

Architectures	Settings	Output shape
Graph-to-Seq Attention layer	heads=8	(B,256)
	embed_dim=256	
	dropout=0.5	
Seq-to-Graph Attention layer	heads=8	(B,256)
	embed_dim=256	
	dropout=0.5	
Concatenation layer	dim=1	(B,512)
Fully-connected Classifier (3 layers)	FC ₁ : 512 $\rightarrow$ 512, GELU, dropout = 0.5	(B,2)
	FC ₂ : 512 $\rightarrow$ 128, GELU, dropout = 0.5	
	FC ₃ : 128 $\rightarrow$ 2	

Note: B represents the batch size in the output tensor shape.

Table S9: The predicted results of DeepCTMS model on 165 ChIP-seq datasets

Dataset	Accuracy	ROC-AUC	PR-AUC
wgEncodeAwgTfbsBroadDnd41CtcfUniPk	0.938244854	0.979842319	0.982335791
wgEncodeAwgTfbsBroadDnd41Ezh239875UniPk	0.681073025	0.722616633	0.746194129
wgEncodeAwgTfbsBroadGm12878CtcfUniPk	0.919811049	0.969662639	0.974733854
wgEncodeAwgTfbsBroadH1hescCtcfUniPk	0.940296193	0.982077631	0.983176469
wgEncodeAwgTfbsBroadHela3CtcfUniPk	0.903420027	0.959800309	0.966564838
wgEncodeAwgTfbsBroadHela3Ezh239875UniPk	0.629577465	0.666507645	0.688002994
wgEncodeAwgTfbsBroadHela3Pol2bUniPk	0.665271967	0.715565877	0.711253279
wgEncodeAwgTfbsBroadHepg2CtcfUniPk	0.943147882	0.981135725	0.983318456
wgEncodeAwgTfbsBroadHepg2Ezh239875UniPk	0.685255921	0.742636039	0.730988837
wgEncodeAwgTfbsBroadHmecCtcfUniPk	0.949906155	0.985958169	0.988300017
wgEncodeAwgTfbsBroadHsmmCtcfUniPk	0.935475709	0.981190542	0.982556603
wgEncodeAwgTfbsBroadHsmmEzh239875UniPk	0.652597403	0.681822496	0.656390764
wgEncodeAwgTfbsBroadHsmmtCtcfUniPk	0.926298248	0.973834378	0.977712438
wgEncodeAwgTfbsBroadHuvecCtcfUniPk	0.938042131	0.979479829	0.982917867
wgEncodeAwgTfbsBroadHuvecEzh239875UniPk	0.704946996	0.781104674	0.777258146
wgEncodeAwgTfbsBroadHuvecPol2bUniPk	0.690019194	0.759606915	0.748357153
wgEncodeAwgTfbsBroadK562CtcfUniPk	0.890823178	0.950091544	0.960465874
wgEncodeAwgTfbsBroadK562Ezh239875UniPk	0.626488095	0.643402778	0.647234878
wgEncodeAwgTfbsBroadK562Hdac2a300705aUniPk	0.703244275	0.760920495	0.75393973
wgEncodeAwgTfbsBroadK562Hdac6a301341aUniPk	0.660674157	0.69280055	0.695007795
wgEncodeAwgTfbsBroadNhaCtcfUniPk	0.942134907	0.979063232	0.981840439
wgEncodeAwgTfbsBroadNhdfadCtcfUniPk	0.939988398	0.981760928	0.983565755
wgEncodeAwgTfbsBroadNhekCtcfUniPk	0.924008992	0.970954965	0.97454672
wgEncodeAwgTfbsBroadNhekPol2bUniPk	0.669761273	0.726115461	0.721226787
wgEncodeAwgTfbsBroadNhlfCtcfUniPk	0.92905119	0.973058447	0.976488231
wgEncodeAwgTfbsBroadOsteoblCtcfUniPk	0.913408859	0.967749584	0.974038944
wgEncodeAwgTfbsHaibA549Elf1V0422111Etoh02UniPk	0.814726841	0.886791208	0.899212849
wgEncodeAwgTfbsHaibA549Fosl2V0422111Etoh02UniPk	0.891759028	0.956239425	0.962500211
wgEncodeAwgTfbsHaibA549Foxa1V0416102Dex100nmUniPk	0.844061048	0.916477337	0.921560229
wgEncodeAwgTfbsHaibA549GabpV0422111Etoh02UniPk	0.814588338	0.886954505	0.899662
wgEncodeAwgTfbsHaibA549Sin3ak20V0422111Etoh02UniPk	0.697478992	0.759389316	0.760396318
wgEncodeAwgTfbsHaibA549Taf1V0422111Etoh02UniPk	0.711700229	0.792604006	0.798883826
wgEncodeAwgTfbsHaibEcc1Foxa1sc6553V0416102Dm002p1hUniPk	0.853511417	0.919626047	0.926584589
wgEncodeAwgTfbsHaibGm12878Elf1sc631V0416101UniPk	0.818202197	0.898459199	0.904723084
wgEncodeAwgTfbsHaibGm12878GabpPcr2xUniPk	0.845671759	0.923768236	0.930935823
wgEncodeAwgTfbsHaibGm12878P300Pcr1xUniPk	0.77408143	0.849189091	0.853291204
wgEncodeAwgTfbsHaibGm12878Pax5c20Pcr1xUniPk	0.832245758	0.910788971	0.913293374
wgEncodeAwgTfbsHaibGm12878Pax5n19Pcr1xUniPk	0.810033531	0.891573538	0.899264774
wgEncodeAwgTfbsHaibGm12878Taf1Pcr1xUniPk	0.769948001	0.851413179	0.858930624
wgEncodeAwgTfbsHaibGm12878Yy1sc281Pcr1xUniPk	0.847374543	0.924277509	0.934971554
wgEncodeAwgTfbsHaibGm12891Pax5c20V0416101UniPk	0.781526104	0.866349206	0.876257766
wgEncodeAwgTfbsHaibGm12892Pax5c20V0416101UniPk	0.76154224	0.845505731	0.854979189
wgEncodeAwgTfbsHaibGm12892Taf1V0416102UniPk	0.760422306	0.841000363	0.851884797
wgEncodeAwgTfbsHaibGm12892Yy1V0416101UniPk	0.841705702	0.921612252	0.931035484
wgEncodeAwgTfbsHaibH1hescFosl1sc183V0416102UniPk	0.752272727	0.816695602	0.827300771
wgEncodeAwgTfbsHaibH1hescGabpPcr1xUniPk	0.820695807	0.89675435	0.906234575

wgEncodeAwgTfbsHaibH1heschdac2sc6296V0416102UniPk	0.762840554	0.843671704	0.842844906
wgEncodeAwgTfbsHaibH1hescSp1Pcr1xUniPk	0.793978349	0.880811423	0.88921095
wgEncodeAwgTfbsHaibH1hescSp2V0422111UniPk	0.838204593	0.908996213	0.92425261
wgEncodeAwgTfbsHaibH1hescSrfPcr1xUniPk	0.863205192	0.934748103	0.944334227
wgEncodeAwgTfbsHaibH1hesctead4sc101184V0422111UniPk	0.897682162	0.958290521	0.960731624
wgEncodeAwgTfbsHaibH1hescYy1sc281V0416102UniPk	0.850027902	0.928450639	0.934750261
wgEncodeAwgTfbsHaibHct116Yy1sc281V0416101UniPk	0.83675913	0.910365466	0.919715015
wgEncodeAwgTfbsHaibHelas3GabpPcr1xUniPk	0.86242378	0.930268528	0.939332485
wgEncodeAwgTfbsHaibHepg2Cebpbsc150V0416101UniPk	0.930787589	0.978947636	0.978739592
wgEncodeAwgTfbsHaibHepg2Elf1sc631V0416101UniPk	0.867777935	0.937968946	0.941470744
wgEncodeAwgTfbsHaibHepg2Fosl2V0416101UniPk	0.866935484	0.937104114	0.943224547
wgEncodeAwgTfbsHaibHepg2Foxa1sc101058V0416101UniPk	0.899928944	0.961409911	0.961920987
wgEncodeAwgTfbsHaibHepg2Foxa1sc6553V0416101UniPk	0.905065816	0.963882601	0.964745418
wgEncodeAwgTfbsHaibHepg2GabpPcr2xUniPk	0.829175153	0.904262725	0.910500008
wgEncodeAwgTfbsHaibHepg2Hdac2sc6296V0416101UniPk	0.817195661	0.897438227	0.902667737
wgEncodeAwgTfbsHaibHepg2JundPcr1xUniPk	0.848172446	0.927091166	0.934683421
wgEncodeAwgTfbsHaibHepg2P300V0416101UniPk	0.854450645	0.930327681	0.931638099
wgEncodeAwgTfbsHaibHepg2Sin3ak20Pcr1xUniPk	0.761860894	0.839849356	0.846064579
wgEncodeAwgTfbsHaibHepg2Sp1Pcr1xUniPk	0.84020825	0.917129821	0.923991026
wgEncodeAwgTfbsHaibHepg2Sp2V0422111UniPk	0.88150289	0.937069222	0.94706665
wgEncodeAwgTfbsHaibHepg2SrfV0416101UniPk	0.842406877	0.915098274	0.923762458
wgEncodeAwgTfbsHaibHepg2Taf1Pcr2xUniPk	0.754374901	0.837108994	0.839211399
wgEncodeAwgTfbsHaibHepg2Tead4sc101184V0422111UniPk	0.833836351	0.909497626	0.915469472
wgEncodeAwgTfbsHaibHepg2Yy1sc281V0416101UniPk	0.803320928	0.884208974	0.88701783
wgEncodeAwgTfbsHaibK562Cebpbsc150V0422111UniPk	0.896051571	0.958575176	0.962408701
wgEncodeAwgTfbsHaibK562Elf1sc631V0416102UniPk	0.887291494	0.951840055	0.952257349
wgEncodeAwgTfbsHaibK562Fosl1sc183V0416101UniPk	0.918506715	0.96987412	0.975481957
wgEncodeAwgTfbsHaibK562GabpV0416101UniPk	0.848132928	0.920227244	0.925782509
wgEncodeAwgTfbsHaibK562Gata2sc267Pcr1xUniPk	0.888844569	0.954513819	0.954771589
wgEncodeAwgTfbsHaibK562Hdac2sc6296V0416102UniPk	0.775711575	0.857067044	0.858203341
wgEncodeAwgTfbsHaibK562MaxV0416102UniPk	0.827381969	0.906165435	0.914786173
wgEncodeAwgTfbsHaibK562Sin3ak20V0416101UniPk	0.774954991	0.857435944	0.859621808
wgEncodeAwgTfbsHaibK562SrfV0416101UniPk	0.81175836	0.886690997	0.901108556
wgEncodeAwgTfbsHaibK562Taf1V0416101UniPk	0.771423694	0.852644227	0.858182722
wgEncodeAwgTfbsHaibK562Tead4sc101184V0422111UniPk	0.843889532	0.921291035	0.924808906
wgEncodeAwgTfbsHaibK562Yy1V0416101UniPk	0.831369362	0.907280706	0.910400954
wgEncodeAwgTfbsHaibK562Yy1V0416102UniPk	0.841377091	0.917944367	0.925745083
wgEncodeAwgTfbsHaibPanc1Sin3ak20V0416101UniPk	0.720864329	0.798366748	0.791917771
wgEncodeAwgTfbsHaibSknsrap300V0416102UniPk	0.817972855	0.899633282	0.903155547
wgEncodeAwgTfbsHaibSknsraYy1sc281V0416102UniPk	0.823616115	0.902178518	0.910701885
wgEncodeAwgTfbsHaibSknshtaf1V0416101UniPk	0.770719241	0.858029203	0.855898054
wgEncodeAwgTfbsHaibT47dFoxa1sc6553V0416102Dm002p1hUniPk	0.887404229	0.953981541	0.955508057
wgEncodeAwgTfbsHaibT47dGata3sc268V0416102Dm002p1hUniPk	0.84266502	0.921220095	0.925995224
wgEncodeAwgTfbsHaibT47dP300V0416102Dm002p1hUniPk	0.793550337	0.875914491	0.878865065
wgEncodeAwgTfbsSydhA549CebpIggrabUniPk	0.940277869	0.982896522	0.982709311
wgEncodeAwgTfbsSydhGm10847NfkbTnfalIggrabUniPk	0.801209788	0.882037296	0.897334054
wgEncodeAwgTfbsSydhGm12878CfosUniPk	0.916284404	0.969211505	0.973951486
wgEncodeAwgTfbsSydhGm12878JundUniPk	0.887063655	0.953112507	0.944728076
wgEncodeAwgTfbsSydhGm12878MaxIggnusUniPk	0.750658295	0.84024984	0.848607631

wgEncodeAwgTfbsSydhGm12878NfkbTnfaIggrabUniPk	0.823547034	0.90496498	0.913296038
wgEncodeAwgTfbsSydhGm12878P300bUniPk	0.854020619	0.927518303	0.929402657
wgEncodeAwgTfbsSydhGm12878Stat3IggmusUniPk	0.7133883	0.77975816	0.762465385
wgEncodeAwgTfbsSydhGm12878TbpIggmusUniPk	0.726804124	0.80803631	0.808148451
wgEncodeAwgTfbsSydhGm12891NfkbTnfaIggrabUniPk	0.823947415	0.90177713	0.910685238
wgEncodeAwgTfbsSydhGm12892NfkbTnfaIggrabUniPk	0.804787069	0.889031966	0.899828688
wgEncodeAwgTfbsSydhGm18526NfkbTnfaIggrabUniPk	0.798507463	0.878885697	0.892954224
wgEncodeAwgTfbsSydhGm19099NfkbTnfaIggrabUniPk	0.847440945	0.918645988	0.924937968
wgEncodeAwgTfbsSydhH1hesCebpbIggrabUniPk	0.940013076	0.980297713	0.981378624
wgEncodeAwgTfbsSydhH1hesCtbp2UcdUniPk	0.727564103	0.797981325	0.795698623
wgEncodeAwgTfbsSydhH1hesCJundIggrabUniPk	0.853528628	0.92416513	0.930827272
wgEncodeAwgTfbsSydhH1hesCMaxUcdUniPk	0.835725678	0.914782654	0.9252514
wgEncodeAwgTfbsSydhH1hesCSin3anb6001263IggrabUniPk	0.774005935	0.853333109	0.862854182
wgEncodeAwgTfbsSydhHela3CebpbIggrabUniPk	0.89908953	0.962117286	0.962821607
wgEncodeAwgTfbsSydhHela3CfosUniPk	0.908664459	0.961285775	0.964413256
wgEncodeAwgTfbsSydhHela3CjunIggrabUniPk	0.895006218	0.963698952	0.966689938
wgEncodeAwgTfbsSydhHela3Hae2f1UniPk	0.804902451	0.887083249	0.885005964
wgEncodeAwgTfbsSydhHela3JundIggrabUniPk	0.909275669	0.96426356	0.968008247
wgEncodeAwgTfbsSydhHela3MaxIggrabUniPk	0.799532589	0.882533307	0.888705136
wgEncodeAwgTfbsSydhHela3P300sc584sc584IggrabUniPk	0.862904828	0.932760726	0.935303541
wgEncodeAwgTfbsSydhHela3Stat3IggrabUniPk	0.832313114	0.906535079	0.915614615
wgEncodeAwgTfbsSydhHepg2CebpbForsklnUniPk	0.909842845	0.967388693	0.969461768
wgEncodeAwgTfbsSydhHepg2CebpbIggrabUniPk	0.951075003	0.987613691	0.986390925
wgEncodeAwgTfbsSydhHepg2CjunIggrabUniPk	0.931337325	0.977064453	0.980014141
wgEncodeAwgTfbsSydhHepg2Corestsc30189IggrabUniPk	0.723950211	0.792063288	0.790943713
wgEncodeAwgTfbsSydhHepg2JundIggrabUniPk	0.942610063	0.980936423	0.982734822
wgEncodeAwgTfbsSydhHepg2MaxIggrabUniPk	0.804644227	0.886499743	0.896870929
wgEncodeAwgTfbsSydhHepg2TbpIggrabUniPk	0.745675377	0.830294142	0.833558922
wgEncodeAwgTfbsSydhHuvecCfosUcdUniPk	0.904007456	0.962739697	0.966301137
wgEncodeAwgTfbsSydhHuvecCjunUniPk	0.920915713	0.973255375	0.974752553
wgEncodeAwgTfbsSydhHuvecGata2UcdUniPk	0.830475119	0.907489062	0.91267871
wgEncodeAwgTfbsSydhHuvecMaxUniPk	0.861504302	0.935076042	0.940436617
wgEncodeAwgTfbsSydhImr90CebpbIggrabUniPk	0.929816246	0.978394818	0.977417367
wgEncodeAwgTfbsSydhK562CebpbIggrabUniPk	0.937764831	0.981185207	0.981396214
wgEncodeAwgTfbsSydhK562CfosUniPk	0.933823529	0.977556227	0.979663883
wgEncodeAwgTfbsSydhK562CjunUniPk	0.915416451	0.96681855	0.969365022
wgEncodeAwgTfbsSydhK562Corestsc30189IggrabUniPk	0.83967836	0.919201014	0.924371559
wgEncodeAwgTfbsSydhK562Gata2UcdUniPk	0.874369445	0.943658832	0.947027574
wgEncodeAwgTfbsSydhK562JundIggrabUniPk	0.882529161	0.950057479	0.954460958
wgEncodeAwgTfbsSydhK562MaxIggrabUniPk	0.821095335	0.902409336	0.907394154
wgEncodeAwgTfbsSydhK562P300IggrabUniPk	0.848224003	0.924772793	0.930654576
wgEncodeAwgTfbsSydhK562TbpIggmusUniPk	0.770074594	0.847193774	0.857659615
wgEncodeAwgTfbsSydhK562Yy1UcdUniPk	0.908900524	0.960002237	0.967232588
wgEncodeAwgTfbsSydhMcf10aesCfosTam112hHvdUniPk	0.926039546	0.977172444	0.976988413
wgEncodeAwgTfbsSydhMcf10aesCfosTam14hHvdUniPk	0.92073389	0.974091128	0.974056617
wgEncodeAwgTfbsSydhMcf10aesCfosTamHvdUniPk	0.930236807	0.978566201	0.978550419
wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01bUniPk	0.899846617	0.964487735	0.965044946
wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01cUniPk	0.90363535	0.965210171	0.967265016
wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01UniPk	0.879463364	0.949127764	0.949864281

wgEncodeAwgTfbsSydhMcf10aesStat3Tam112hHvdUniPk	0.903385366	0.964423911	0.964508149
wgEncodeAwgTfbsSydhMcf10aesStat3TamUniPk	0.903766174	0.966186617	0.96721173
wgEncodeAwgTfbsSydhMcf7Gata3UcdUniPk	0.738840217	0.817760279	0.825586536
wgEncodeAwgTfbsSydhMcf7Hae2f1UcdUniPk	0.761073476	0.846028876	0.848384672
wgEncodeAwgTfbsSydhNb4CmycUniPk	0.811597013	0.891449537	0.899663695
wgEncodeAwgTfbsSydhNb4MaxUniPk	0.835488611	0.915115739	0.92222883
wgEncodeAwgTfbsSydhNt2d1Yy1UcdUniPk	0.886448848	0.947972438	0.958286795
wgEncodeAwgTfbsSydhShsy5yGata2UcdUniPk	0.857766018	0.932689791	0.939159018
wgEncodeAwgTfbsSydhShsy5yGata3sc269sc269UcdUniPk	0.826066047	0.907616527	0.914292386
wgEncodeAwgTfbsUchicagoK562EfosUniPk	0.933746898	0.97290435	0.97786723
wgEncodeAwgTfbsUchicagoK562Egata2UniPk	0.788982674	0.865200766	0.875915122
wgEncodeAwgTfbsUchicagoK562Ehdac8UniPk	0.609649123	0.649983317	0.651924804
wgEncodeAwgTfbsUchicagoK562EjundUniPk	0.848032175	0.922753039	0.932791608
wgEncodeAwgTfbsUtaGm12878CmycUniPk	0.729080933	0.798785043	0.789938136
wgEncodeAwgTfbsUtaGm12878CtcfUniPk	0.951916967	0.985791413	0.986897351
wgEncodeAwgTfbsUtaH1hesCmycUniPk	0.632989691	0.645857425	0.623722268
wgEncodeAwgTfbsUtaH1hesCtcfUniPk	0.941482752	0.982555959	0.982618877
wgEncodeAwgTfbsUtaHelas3CtcfUniPk	0.95695612	0.987293192	0.987160311
wgEncodeAwgTfbsUtaHepg2CmycUniPk	0.789931585	0.857649912	0.861308417
wgEncodeAwgTfbsUtaK562CtcfUniPk	0.955994839	0.988598867	0.989214392
wgEncodeAwgTfbsUtaMcf7CtcfUniPk	0.917158975	0.968807619	0.970049539
Average	0.837094712	0.902132217	0.906037431

Table S10: The predicted results of MSGSNet model on 165 ChIP-seq datasets

Dataset	Accuracy	ROC-AUC	PR-AUC
wgEncodeAwgTfbsBroadDnd41CtcfUniPk	0.912397	0.974698	0.976768
wgEncodeAwgTfbsBroadDnd41Ezh239875UniPk	0.700447	0.762464	0.771877
wgEncodeAwgTfbsBroadGm12878CtcfUniPk	0.90731	0.968477	0.972392
wgEncodeAwgTfbsBroadH1hescCtcfUniPk	0.905637	0.975101	0.975477
wgEncodeAwgTfbsBroadHela3CtcfUniPk	0.893352	0.960232	0.966255
wgEncodeAwgTfbsBroadHela3Ezh239875UniPk	0.647887	0.70612	0.726512
wgEncodeAwgTfbsBroadHela3Pol2bUniPk	0.676987	0.753172	0.748059
wgEncodeAwgTfbsBroadHepg2CtcfUniPk	0.913284	0.978023	0.979679
wgEncodeAwgTfbsBroadHepg2Ezh239875UniPk	0.721925	0.793947	0.767475
wgEncodeAwgTfbsBroadHmecCtcfUniPk	0.925118	0.980729	0.982134
wgEncodeAwgTfbsBroadHsmmCtcfUniPk	0.890435	0.975907	0.976731
wgEncodeAwgTfbsBroadHsmmEzh239875UniPk	0.660714	0.70319	0.668089
wgEncodeAwgTfbsBroadHsmmtCtcfUniPk	0.904258	0.971667	0.973159
wgEncodeAwgTfbsBroadHuvecCtcfUniPk	0.914429	0.975956	0.978446
wgEncodeAwgTfbsBroadHuvecEzh239875UniPk	0.716726	0.803106	0.804843
wgEncodeAwgTfbsBroadHuvecPol2bUniPk	0.726488	0.800381	0.787798
wgEncodeAwgTfbsBroadK562CtcfUniPk	0.884606	0.953353	0.961219
wgEncodeAwgTfbsBroadK562Ezh239875UniPk	0.644345	0.69903	0.62914
wgEncodeAwgTfbsBroadK562Hdac2a300705aUniPk	0.720897	0.801691	0.805268
wgEncodeAwgTfbsBroadK562Hdac6a301341aUniPk	0.626966	0.704806	0.709993
wgEncodeAwgTfbsBroadNhaCtcfUniPk	0.92202	0.97595	0.977776
wgEncodeAwgTfbsBroadNhdfadCtcfUniPk	0.899172	0.974524	0.973209
wgEncodeAwgTfbsBroadNhekCtcfUniPk	0.893525	0.969552	0.972501
wgEncodeAwgTfbsBroadNhekPol2bUniPk	0.677719	0.751425	0.749901
wgEncodeAwgTfbsBroadNhlfCtcfUniPk	0.923443	0.974717	0.97769
wgEncodeAwgTfbsBroadOsteoblCtcfUniPk	0.88323	0.961492	0.967348
wgEncodeAwgTfbsHaibA549Elf1V0422111Etoh02UniPk	0.817399	0.904782	0.909895
wgEncodeAwgTfbsHaibA549Fosl2V0422111Etoh02UniPk	0.891938	0.956908	0.960103
wgEncodeAwgTfbsHaibA549Foxa1V0416102Dex100nmUniPk	0.860982	0.929083	0.934207
wgEncodeAwgTfbsHaibA549GabpV0422111Etoh02UniPk	0.812841	0.90233	0.91006
wgEncodeAwgTfbsHaibA549Sin3ak20V0422111Etoh02UniPk	0.726471	0.804083	0.808531
wgEncodeAwgTfbsHaibA549Taf1V0422111Etoh02UniPk	0.72623	0.815959	0.822799
wgEncodeAwgTfbsHaibEcc1Foxa1sc6553V0416102Dm002p1hUniPk	0.850495	0.926996	0.928529
wgEncodeAwgTfbsHaibGm12878Elf1sc631V0416101UniPk	0.820108	0.903759	0.905523
wgEncodeAwgTfbsHaibGm12878GabpPcr2xUniPk	0.851939	0.933264	0.938372
wgEncodeAwgTfbsHaibGm12878P300Pcr1xUniPk	0.779543	0.876261	0.882955
wgEncodeAwgTfbsHaibGm12878Pax5c20Pcr1xUniPk	0.816183	0.898579	0.902884
wgEncodeAwgTfbsHaibGm12878Pax5n19Pcr1xUniPk	0.811194	0.901841	0.906652
wgEncodeAwgTfbsHaibGm12878Taf1Pcr1xUniPk	0.781424	0.867122	0.87265
wgEncodeAwgTfbsHaibGm12878Yy1sc281Pcr1xUniPk	0.835909	0.929876	0.938006
wgEncodeAwgTfbsHaibGm12891Pax5c20V0416101UniPk	0.793574	0.88012	0.88673
wgEncodeAwgTfbsHaibGm12892Pax5c20V0416101UniPk	0.784627	0.867585	0.873296
wgEncodeAwgTfbsHaibGm12892Taf1V0416102UniPk	0.772604	0.859453	0.867646
wgEncodeAwgTfbsHaibGm12892Yy1V0416101UniPk	0.844129	0.930446	0.935642
wgEncodeAwgTfbsHaibH1hescFosl1sc183V0416102UniPk	0.754545	0.826864	0.849366
wgEncodeAwgTfbsHaibH1hescGabpPcr1xUniPk	0.82694	0.911702	0.916935

wgEncodeAwgTfbsHaibH1heschdac2sc6296V0416102UniPk	0.782046	0.862108	0.865996
wgEncodeAwgTfbsHaibH1heschSp1Pcr1xUniPk	0.794317	0.90266	0.910305
wgEncodeAwgTfbsHaibH1heschSp2V0422111UniPk	0.845511	0.930851	0.938718
wgEncodeAwgTfbsHaibH1heschSrfPcr1xUniPk	0.876186	0.949455	0.956667
wgEncodeAwgTfbsHaibH1heschTead4sc101184V0422111UniPk	0.888462	0.960972	0.960591
wgEncodeAwgTfbsHaibH1heschYy1sc281V0416102UniPk	0.857701	0.937434	0.941486
wgEncodeAwgTfbsHaibHct116Yy1sc281V0416101UniPk	0.834963	0.915524	0.924466
wgEncodeAwgTfbsHaibHela3GbpPcr1xUniPk	0.876143	0.944484	0.952599
wgEncodeAwgTfbsHaibHepg2Cebpssc150V0416101UniPk	0.915064	0.978618	0.976983
wgEncodeAwgTfbsHaibHepg2Elf1sc631V0416101UniPk	0.867213	0.952239	0.950416
wgEncodeAwgTfbsHaibHepg2Fosl2V0416101UniPk	0.872077	0.943642	0.946745
wgEncodeAwgTfbsHaibHepg2Foxa1sc101058V0416101UniPk	0.892527	0.961059	0.960122
wgEncodeAwgTfbsHaibHepg2Foxa1sc6553V0416101UniPk	0.910052	0.966576	0.966639
wgEncodeAwgTfbsHaibHepg2GbpPcr2xUniPk	0.852088	0.922438	0.929421
wgEncodeAwgTfbsHaibHepg2Hdac2sc6296V0416101UniPk	0.835275	0.911109	0.913606
wgEncodeAwgTfbsHaibHepg2JundPcr1xUniPk	0.859536	0.936985	0.940869
wgEncodeAwgTfbsHaibHepg2P300V0416101UniPk	0.848779	0.93528	0.936042
wgEncodeAwgTfbsHaibHepg2Sin3ak20Pcr1xUniPk	0.78136	0.861487	0.867375
wgEncodeAwgTfbsHaibHepg2Sp1Pcr1xUniPk	0.839808	0.929219	0.934037
wgEncodeAwgTfbsHaibHepg2Sp2V0422111UniPk	0.872832	0.939164	0.95186
wgEncodeAwgTfbsHaibHepg2SrfV0416101UniPk	0.839542	0.926262	0.93207
wgEncodeAwgTfbsHaibHepg2Taf1Pcr2xUniPk	0.764622	0.857082	0.861965
wgEncodeAwgTfbsHaibHepg2Tead4sc101184V0422111UniPk	0.842052	0.917571	0.922785
wgEncodeAwgTfbsHaibHepg2Yy1sc281V0416101UniPk	0.817635	0.898492	0.900073
wgEncodeAwgTfbsHaibK562Cebpssc150V0422111UniPk	0.886382	0.960431	0.961132
wgEncodeAwgTfbsHaibK562Elf1sc631V0416102UniPk	0.876417	0.953758	0.94924
wgEncodeAwgTfbsHaibK562Fosl1sc183V0416101UniPk	0.917596	0.975405	0.978852
wgEncodeAwgTfbsHaibK562GbpV0416101UniPk	0.867965	0.936126	0.939271
wgEncodeAwgTfbsHaibK562Gata2sc267Pcr1xUniPk	0.884191	0.954715	0.955932
wgEncodeAwgTfbsHaibK562Hdac2sc6296V0416102UniPk	0.787476	0.885104	0.890941
wgEncodeAwgTfbsHaibK562MaxV0416102UniPk	0.839057	0.915802	0.920543
wgEncodeAwgTfbsHaibK562Sin3ak20V0416101UniPk	0.777155	0.86729	0.872196
wgEncodeAwgTfbsHaibK562SrfV0416101UniPk	0.819849	0.903181	0.910982
wgEncodeAwgTfbsHaibK562Taf1V0416101UniPk	0.785592	0.875451	0.882373
wgEncodeAwgTfbsHaibK562Tead4sc101184V0422111UniPk	0.851083	0.924454	0.926406
wgEncodeAwgTfbsHaibK562Yy1V0416101UniPk	0.828119	0.917433	0.923267
wgEncodeAwgTfbsHaibK562Yy1V0416102UniPk	0.834299	0.926891	0.932832
wgEncodeAwgTfbsHaibPanc1Sin3ak20V0416101UniPk	0.739639	0.827524	0.829357
wgEncodeAwgTfbsHaibSknsrap300V0416102UniPk	0.829411	0.912525	0.916299
wgEncodeAwgTfbsHaibSknsraYy1sc281V0416102UniPk	0.82378	0.910165	0.915332
wgEncodeAwgTfbsHaibSknsHTaf1V0416101UniPk	0.797919	0.87999	0.88301
wgEncodeAwgTfbsHaibT47dFoxa1sc6553V0416102Dm002p1hUniPk	0.880785	0.956416	0.957867
wgEncodeAwgTfbsHaibT47dGata3sc268V0416102Dm002p1hUniPk	0.842802	0.91924	0.921244
wgEncodeAwgTfbsHaibT47dP300V0416102Dm002p1hUniPk	0.797561	0.896046	0.899297
wgEncodeAwgTfbsSydhA549CebpIggrabUniPk	0.935669	0.982628	0.981866
wgEncodeAwgTfbsSydhGm10847NfkbTnfalIggrabUniPk	0.815232	0.897369	0.903547
wgEncodeAwgTfbsSydhGm12878CfosUniPk	0.917431	0.979355	0.98231
wgEncodeAwgTfbsSydhGm12878JundUniPk	0.87577	0.951934	0.948865
wgEncodeAwgTfbsSydhGm12878MaxIggmusUniPk	0.772736	0.852739	0.85547

wgEncodeAwgTfbsSydhGm12878NfkbTnfaIggrabUniPk	0.836429	0.917333	0.922807
wgEncodeAwgTfbsSydhGm12878P300bUniPk	0.866804	0.937104	0.941011
wgEncodeAwgTfbsSydhGm12878Stat3IggmusUniPk	0.729093	0.811213	0.807754
wgEncodeAwgTfbsSydhGm12878TbpIggmusUniPk	0.753265	0.838935	0.837576
wgEncodeAwgTfbsSydhGm12891NfkbTnfaIggrabUniPk	0.83434	0.91439	0.919816
wgEncodeAwgTfbsSydhGm12892NfkbTnfaIggrabUniPk	0.830588	0.910241	0.918892
wgEncodeAwgTfbsSydhGm18526NfkbTnfaIggrabUniPk	0.786899	0.898196	0.91453
wgEncodeAwgTfbsSydhGm19099NfkbTnfaIggrabUniPk	0.847769	0.934437	0.939718
wgEncodeAwgTfbsSydhH1hesCebpbIggrabUniPk	0.932821	0.98176	0.98087
wgEncodeAwgTfbsSydhH1hesCtbp2UcdUniPk	0.757479	0.833616	0.83727
wgEncodeAwgTfbsSydhH1hesCJundIggrabUniPk	0.854061	0.934213	0.941667
wgEncodeAwgTfbsSydhH1hesCMaxUcdUniPk	0.845067	0.924718	0.93
wgEncodeAwgTfbsSydhH1hesCSin3anb6001263IggrabUniPk	0.782671	0.866379	0.873267
wgEncodeAwgTfbsSydhHela3CebpbIggrabUniPk	0.891123	0.960266	0.95813
wgEncodeAwgTfbsSydhHela3CfosUniPk	0.888245	0.962494	0.96212
wgEncodeAwgTfbsSydhHela3CjunIggrabUniPk	0.898158	0.95903	0.960228
wgEncodeAwgTfbsSydhHela3Hae2f1UniPk	0.82041	0.90818	0.906099
wgEncodeAwgTfbsSydhHela3JundIggrabUniPk	0.907812	0.965581	0.966237
wgEncodeAwgTfbsSydhHela3MaxIggrabUniPk	0.812083	0.895189	0.901501
wgEncodeAwgTfbsSydhHela3P300sc584sc584IggrabUniPk	0.864991	0.939749	0.940864
wgEncodeAwgTfbsSydhHela3Stat3IggrabUniPk	0.837878	0.914167	0.922365
wgEncodeAwgTfbsSydhHepg2CebpbForsklnUniPk	0.907361	0.968245	0.968977
wgEncodeAwgTfbsSydhHepg2CebpbIggrabUniPk	0.940527	0.986248	0.984427
wgEncodeAwgTfbsSydhHepg2CjunIggrabUniPk	0.926547	0.978077	0.980275
wgEncodeAwgTfbsSydhHepg2Corestsc30189IggrabUniPk	0.743542	0.820916	0.826952
wgEncodeAwgTfbsSydhHepg2JundIggrabUniPk	0.945755	0.983881	0.984281
wgEncodeAwgTfbsSydhHepg2MaxIggrabUniPk	0.826374	0.905451	0.90782
wgEncodeAwgTfbsSydhHepg2TbpIggrabUniPk	0.774199	0.854441	0.857757
wgEncodeAwgTfbsSydhHuvecCfosUcdUniPk	0.904994	0.965828	0.968947
wgEncodeAwgTfbsSydhHuvecCjunUniPk	0.920482	0.975684	0.974354
wgEncodeAwgTfbsSydhHuvecGata2UcdUniPk	0.841814	0.921908	0.92731
wgEncodeAwgTfbsSydhHuvecMaxUniPk	0.855676	0.944387	0.947619
wgEncodeAwgTfbsSydhImr90CebpbIggrabUniPk	0.923254	0.975245	0.971746
wgEncodeAwgTfbsSydhK562CebpbIggrabUniPk	0.908766	0.97245	0.965685
wgEncodeAwgTfbsSydhK562CfosUniPk	0.930816	0.979467	0.979037
wgEncodeAwgTfbsSydhK562CjunUniPk	0.92059	0.971292	0.973579
wgEncodeAwgTfbsSydhK562Corestsc30189IggrabUniPk	0.844873	0.924568	0.927819
wgEncodeAwgTfbsSydhK562Gata2UcdUniPk	0.869325	0.941243	0.941089
wgEncodeAwgTfbsSydhK562JundIggrabUniPk	0.876346	0.950926	0.950058
wgEncodeAwgTfbsSydhK562MaxIggrabUniPk	0.827099	0.907351	0.911752
wgEncodeAwgTfbsSydhK562P300IggrabUniPk	0.847531	0.925261	0.927595
wgEncodeAwgTfbsSydhK562TbpIggmusUniPk	0.784408	0.867948	0.88092
wgEncodeAwgTfbsSydhK562Yy1UcdUniPk	0.901571	0.963448	0.968352
wgEncodeAwgTfbsSydhMcf10aesCfosTam112hHvdUniPk	0.91137	0.975145	0.972427
wgEncodeAwgTfbsSydhMcf10aesCfosTam14hHvdUniPk	0.917064	0.972894	0.97154
wgEncodeAwgTfbsSydhMcf10aesCfosTamHvdUniPk	0.934988	0.980363	0.979667
wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01bUniPk	0.897858	0.963957	0.964149
wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01cUniPk	0.882641	0.966239	0.96623
wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01UniPk	0.872033	0.951779	0.950557

wgEncodeAwgTfbsSydhMcf10aesStat3Tam112hHvdUniPk	0.900623	0.964924	0.962965
wgEncodeAwgTfbsSydhMcf10aesStat3TamUniPk	0.903939	0.967288	0.96577
wgEncodeAwgTfbsSydhMcf7Gata3UcdUniPk	0.764706	0.843271	0.844494
wgEncodeAwgTfbsSydhMcf7Hae2f1UcdUniPk	0.772929	0.851728	0.847772
wgEncodeAwgTfbsSydhNb4CmycUniPk	0.820033	0.901285	0.909726
wgEncodeAwgTfbsSydhNb4MaxUniPk	0.842395	0.922672	0.92652
wgEncodeAwgTfbsSydhNt2d1Yy1UcdUniPk	0.868238	0.953259	0.96018
wgEncodeAwgTfbsSydhShsy5yGata2UcdUniPk	0.847898	0.923495	0.925087
wgEncodeAwgTfbsSydhShsy5yGata3sc269sc269UcdUniPk	0.82318	0.90651	0.912645
wgEncodeAwgTfbsUchicagoK562EfosUniPk	0.927047	0.975237	0.977407
wgEncodeAwgTfbsUchicagoK562Egata2UniPk	0.79076	0.875319	0.884038
wgEncodeAwgTfbsUchicagoK562Ehdac8UniPk	0.652047	0.71261	0.705036
wgEncodeAwgTfbsUchicagoK562EjundUniPk	0.84018	0.929135	0.933013
wgEncodeAwgTfbsUtaGm12878CmycUniPk	0.748285	0.815873	0.813181
wgEncodeAwgTfbsUtaGm12878CtcfUniPk	0.933118	0.982929	0.982775
wgEncodeAwgTfbsUtaH1hesecCmycUniPk	0.641237	0.704561	0.688059
wgEncodeAwgTfbsUtaH1hesecCtcfUniPk	0.895568	0.971404	0.97052
wgEncodeAwgTfbsUtaHelas3CtcfUniPk	0.90621	0.981454	0.9796
wgEncodeAwgTfbsUtaHepg2CmycUniPk	0.792474	0.880827	0.885116
wgEncodeAwgTfbsUtaK562CtcfUniPk	0.923279	0.983752	0.983903
wgEncodeAwgTfbsUtaMcf7CtcfUniPk	0.886466	0.965066	0.963913
Average	0.838173	0.91269	0.914825

Table S11: The predicted results of DeepCTMS model and MSGSNet model on different cell lines and 165 ChIP-seq datasets

Cell lines	ACC		ROC-AUC		PR-AUC	
	DeepCTMS	MSGSNet	DeepCTMS	MSGSNet	DeepCTMS	MSGSNet
Hepg2	0.844787	0.850094	0.910834	0.922588	0.914145	0.924376
K562	0.833319	0.833645	0.896283	0.907342	0.900848	0.90809
A549	0.81637	0.824504	0.88305	0.899396	0.889275	0.903923
H1hesc	0.826532	0.827516	0.890592	0.904612	0.894699	0.908575
Helas3	0.840718	0.840098	0.901516	0.911548	0.906067	0.914659
Imr90	0.929816	0.923254	0.978395	0.975245	0.977417	0.971746
Dnd41	0.809659	0.806422	0.851229	0.868581	0.864265	0.874323
Gm12878	0.821773	0.825679	0.893327	0.903917	0.89574	0.90654
Hmec	0.949906	0.925118	0.985958	0.980729	0.9883	0.982134
Hsmm	0.838124	0.818469	0.878949	0.883588	0.87222	0.87266
Huvec	0.835702	0.840087	0.899822	0.912464	0.900386	0.91276
Nha	0.942135	0.92202	0.979063	0.97595	0.98184	0.977776
Nhdfad	0.939988	0.899172	0.981761	0.974524	0.983566	0.973209
Nhek	0.796885	0.785622	0.848535	0.860488	0.847887	0.861201
Nhlf	0.929051	0.923443	0.973058	0.974717	0.976488	0.97769
Osteobl	0.913409	0.88323	0.96775	0.961492	0.974039	0.967348
Ecc1	0.853511	0.850495	0.919626	0.926996	0.926585	0.928529
Gm12891	0.802737	0.813957	0.884063	0.897255	0.893472	0.903273
Gm12892	0.792114	0.807987	0.874288	0.891931	0.884432	0.898869
Hct116	0.836759	0.834963	0.910365	0.915524	0.919715	0.924466
Panc1	0.720864	0.739639	0.798367	0.827524	0.791918	0.829357
Sknshra	0.820794	0.826596	0.900906	0.911345	0.906929	0.915816
T47d	0.841207	0.840383	0.917039	0.923901	0.920123	0.926136
Gm10847	0.80121	0.815232	0.882037	0.897369	0.897334	0.903547
Nb4	0.823543	0.831214	0.903283	0.911979	0.910946	0.918123
Sknsh	0.770719	0.797919	0.858029	0.87999	0.855898	0.88301
Gm18526	0.798507	0.786899	0.878886	0.898196	0.892954	0.91453
Gm19099	0.847441	0.847769	0.918646	0.934437	0.924938	0.939718
Mcf10aes	0.908388	0.902564	0.967408	0.967824	0.967936	0.966663
Nt2d1	0.886449	0.868238	0.947972	0.953259	0.958287	0.96018
Shsy5y	0.841916	0.835539	0.920153	0.915002	0.926726	0.918866
Mcf7	0.805691	0.808034	0.877532	0.886688	0.88134	0.885393

Table S12: The predicted results of DeepCTMS model and MSGSNet model on different transcription factors (TFs) and 165 ChIP-seq datasets

TFs	ACC		ROC-AUC		PR-AUC	
	DeepCTMS	MSGSNet	DeepCTMS	MSGSNet	DeepCTMS	MSGSNet
CEBPB	0.92608	0.915663	0.975269	0.973988	0.975858	0.972202
COREST	0.781814	0.794207	0.855632	0.872742	0.857658	0.877385
CTCF	0.932878	0.905843	0.976435	0.972749	0.979425	0.974575
ELF1	0.847	0.845284	0.918765	0.928635	0.924416	0.928768
EZH2	0.663323	0.682007	0.706348	0.744643	0.707678	0.727989
FOS	0.900251	0.897343	0.954453	0.9578	0.958358	0.96071
FOXA1	0.877994	0.878968	0.943075	0.948026	0.946064	0.949473
GABP	0.836781	0.847986	0.910373	0.925057	0.918741	0.93111
GATA2	0.848088	0.846798	0.92071	0.923336	0.92591	0.926691
GATA3	0.802524	0.810229	0.882199	0.889674	0.888625	0.892794
HAE2F1	0.782988	0.796669	0.866556	0.879954	0.866695	0.876935
HDAC	0.721553	0.734118	0.783647	0.812905	0.784098	0.81514
CJUN	0.915669	0.916444	0.970209	0.971021	0.972705	0.972109
JUND	0.881602	0.879923	0.946054	0.950379	0.949748	0.952141
MAX	0.817004	0.827561	0.897854	0.908539	0.905535	0.912653
CMYC	0.7409	0.750507	0.798435	0.825636	0.793658	0.824021
NFKB	0.816573	0.82521	0.895891	0.911994	0.906506	0.919885
PAX5	0.796337	0.801395	0.878554	0.887031	0.885949	0.89239
POL2B	0.675017	0.693731	0.733763	0.768326	0.726946	0.761919
P300	0.829315	0.833517	0.905731	0.917461	0.908902	0.92058
SIN3A	0.745833	0.761459	0.821675	0.845353	0.824171	0.850145
SP1	0.817093	0.817063	0.898971	0.91594	0.906601	0.922171
SP2	0.859854	0.859171	0.923033	0.935007	0.93566	0.945289
SRF	0.839123	0.845192	0.912179	0.926299	0.923068	0.93324
STAT3	0.862257	0.860581	0.927961	0.934224	0.927425	0.934256
TAF1	0.756431	0.771398	0.8388	0.859176	0.843832	0.865074
TBP	0.74253	0.767338	0.820876	0.848735	0.823766	0.853381
TEAD4	0.858469	0.860532	0.929693	0.934332	0.93367	0.936594
YY1	0.84709	0.844634	0.920429	0.928297	0.927986	0.933964